

The Impervious Paradox: Ridge Regression and SHAP Archetypes of Parcel-Scale Urban Flood Risk

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MAUD '27, Harvard Graduate School of Design

Research submission for AI for Earth and Planetary Science

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Abstract

Urban flood design often assumes that reducing impervious surface area (ISA) will reduce flood risk. This assumption is well supported at the watershed scale, where paving increases runoff and downstream flood volume, but it is less clear whether the same relationship holds at the parcel scale where many urban design interventions are actually applied. This study tests the parcel-scale relationship between imperviousness and flood depth in Houston's Brays Bayou watershed using 115,820 residential parcels and Hurricane Harvey (2017) FEMA flood-depth data as ground truth. I train spatially cross-validated Ridge regression models using twelve public geospatial features, including a custom 10 m Height Above Nearest Drainage (HAND) feature, and interpret model structure using LightGBM SHAP values clustered into residential flood-risk archetypes. The Ridge model achieves a Spearman rank correlation of 0.42 under 2 km block cross-validation, with topographic variables dominating model importance. Across all four residential archetypes, the most impervious parcels flooded less than the least impervious parcels; the inversion is strongest in topographic hotspots, where the most impervious quintile flooded 1.31 ft less than the least impervious quintile. I interpret this impervious paradox as spatial confounding: in mature urban watersheds, low-ISA land often occupies floodplains, drainage easements, parks, and other topographically vulnerable locations. The results suggest that parcel-scale flood-risk screening should treat topographic position as the primary signal and surface cover as a secondary, context-dependent modifier.

Keywords: urban flooding; impervious surface; Ridge regression; SHAP; parcel-scale analysis; Hurricane Harvey; HAND; spatial cross-validation; topographic position

1. Introduction

Urban flood mitigation is increasingly implemented through parcel-scale decisions: which lots to depave, where to install bioswales, which properties to retrofit, and where new development should be allowed or discouraged. A common design prescription is that reducing impervious surface area (ISA) will reduce flood risk (Shuster et al., 2005; Dietz, 2007; Jacobson, 2011). This prescription comes from watershed hydrology, where pavement increases runoff, reduces infiltration, and can intensify downstream flooding.

The watershed-scale relationship is robust, but the parcel-scale design translation is less certain. Flood depth at a parcel is not determined only by the runoff produced on that parcel. It also depends on where the parcel sits in the watershed's hydraulic geometry: its elevation relative to nearby drainage, its distance to streams, local flow accumulation, slope, drainage connectivity, and upstream contributing area. At the parcel scale, impervious surface may therefore function both as a physical runoff variable and as a historical siting proxy.

This distinction matters for urban design. If ISA is treated as the primary parcel-level risk signal, then interventions may prioritize paved parcels even when the deepest flooding occurs on low-lying, less-developed land. Conversely, if topographic position is the dominant signal, then flood-resilience planning should begin with local drainage geometry and use surface cover as a secondary modifier rather than the main criterion.

This study asks a simple research question: does impervious surface predict parcel-level flood depth in the same direction assumed by watershed-scale hydrology? To answer this, I analyze residential parcels in Houston's Brays Bayou watershed using Hurricane Harvey flood depths as ground truth and public geospatial features describing topography, built form, and drainage connectivity.

The paper makes three contributions. First, it shows that a simple, interpretable Ridge regression model can rank parcel-scale flood risk with moderate but useful spatial generalization. Second, it uses SHAP-based clustering to identify four residential flood-risk archetypes, including a small topographic hotspot group that floods much more deeply than the residential baseline. Third, it documents an impervious paradox: within every residential archetype, the most impervious parcels flooded less than the least impervious parcels during Harvey. I argue that this inversion

reflects spatial confounding between surface cover and topographic position rather than a counterintuitive physical benefit of imperviousness.

The rest of the report follows the EPS-210 research-paper structure: Section 2 describes the study area, data, features, and preprocessing; Section 3 describes the modeling, validation, and interpretation methods; Section 4 presents the results; Section 5 discusses physical interpretation, causal limits, and design implications; and Section 6 concludes.

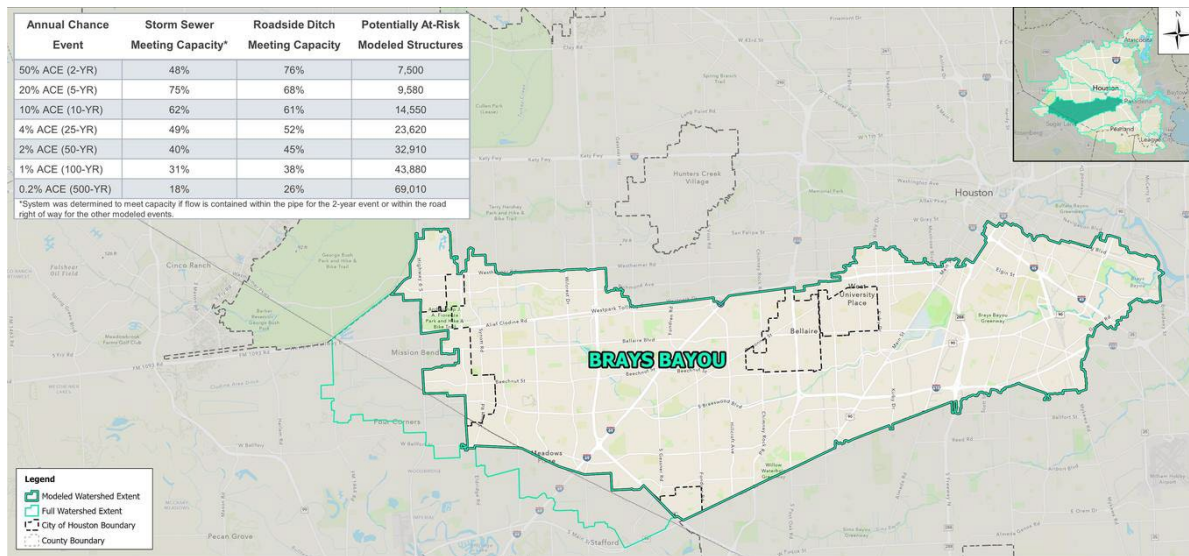


Figure 1. Study area. Brays Bayou in Houston, Texas, showing the modeled watershed extent, full watershed extent, and surrounding jurisdictional context. The inset locates the watershed within the Houston metropolitan area, and the table summarizes drainage-capacity benchmarks and potentially at-risk modeled structures across return periods.

2. Data

2.1 Study area and flood-depth target

The study area is the Brays Bayou watershed in Houston, Texas, a low-relief, heavily urbanized coastal watershed of approximately 330 km². The watershed experienced severe flooding during Hurricane Harvey, which produced roughly 1,000 mm of rainfall over five days in August 2017. This event provides a useful stress test for parcel-scale flood-risk modeling because it generated widespread saturation-limited flooding across an already urbanized landscape.

Ground-truth flood depth is taken from FEMA's Hurricane Harvey post-event analysis raster (FEMA, 2018). For each parcel polygon, I calculate mean and maximum flood depth in feet. The targets are transformed as $\log(1 + \text{depth_ft})$ to reduce the influence of extreme values while preserving rank order, which is the main quantity of interest for intervention targeting.

2.2 Parcel sample and residential filter

The full dataset contains 118,119 parcels from Harris County Appraisal District records intersected with the Brays Bayou watershed and flood-depth raster. Because the research question concerns parcel-scale design decisions for residential land, the headline analysis focuses on residential parcels classified as Developed Low, Medium, or High Intensity in NLCD 2016. Open space, forest, wetlands, water, parks, and easement-like parcels are excluded from the headline residential sample.

The final residential sample contains 115,820 parcels, or 98.1% of the full parcel sample. The exclusion of public and green parcels is important because low-ISA land is often intentionally located in flood-prone positions, such as

drainage reserves or floodplain parks. Removing these parcels tests whether the impervious paradox remains within ordinary residential land rather than being created by designed flood-storage land.

Table 1. Dataset summary.

Item	Value
Watershed	Brays Bayou, Houston, Texas (~330 km ²)
Flood event	Hurricane Harvey, 25-30 August 2017 (~1,000 mm rainfall)
Full parcel sample	118,119 parcels
Headline residential sample	115,820 parcels, excluding public/green land
Feature count	12 public-data predictors
Target variables	Parcel mean and maximum flood depth, $\log(1 + \text{depth_ft})$
Spatial CV schemes	7 hydrologic sub-basin folds; 75 uniform 2 km block folds
Interpretation method	SHAP values from LightGBM clustered with k-means, $k = 4$

2.3 Predictor features

The model uses twelve features derived from public geospatial data. Five features describe topography: elevation, slope, topographic wetness index, log flow accumulation, and distance to stream. These are computed from a 10 m digital elevation model that is hydrologically conditioned before flow routing (Farr et al., 2007).

Four features describe built form and land cover: impervious surface fraction from GHSL 2015, log parcel area, distance to street, and an enclave indicator. Two features describe drainage connectivity: `conn_topo`, the fraction of parcel pixels whose flow path reaches a stream within 500 m, and `Cw_topo`, an ISA-weighted connectivity index defined as $\text{ISA_frac} \times 1 / \max(\text{distance_to_stream}, 1)$. The final feature is a custom 10 m Height Above Nearest Drainage (HAND) variable. I use `HAND_min`, the lowest HAND value inside the parcel, because parcel flood depth is often controlled by the lowest portion of the property rather than its mean elevation.

All continuous variables used in Ridge regression are standardized before fitting. Log transformations are applied to parcel area and flow accumulation. LightGBM is trained on raw values because tree-based models do not require feature standardization.

3. Methods

3.1 Spatial cross-validation

Random parcel hold-out is not appropriate for this problem because nearby parcels share unmeasured conditions such as local drainage infrastructure, neighborhood microtopography, soil drainage, and storm-sewer performance. Random splits would allow the model to exploit spatial autocorrelation and would overstate generalization. I therefore evaluate geographic transfer using spatial cross-validation.

Two complementary split schemes are used. Sub-basin cross-validation divides the watershed into seven hydrologic folds, preserving drainage structure but producing uneven fold sizes. A second 2 km block cross-validation divides the study area into more uniform spatial blocks, reducing dominance by any one large sub-basin. The headline performance results use 2 km block cross-validation because weighted, unweighted, and pooled Spearman values are tightly clustered, suggesting a more stable estimate of geographic generalization.

3.2 Models and evaluation metrics

I compare five models: a median baseline, a single-feature `Cw_topo` linear model, Ridge regression using all twelve features, LightGBM-LambdaRank, and a Random Forest classifier trained on flood-risk classes. Ridge regression is the headline model because it is interpretable, stable under correlated predictors, and performs competitively with the more flexible LightGBM model.

The primary metric is Spearman rank correlation because the main design task is ranking parcels by risk rather than predicting exact flood depth. I also report `NDCG@10%` to evaluate the quality of top-decile risk ranking, which is relevant for targeting limited interventions. For classification comparisons, I report ROC-AUC by class. Exact hyperparameters and full per-fold diagnostics are placed in the appendix to keep the main report readable.

3.3 Model interpretation and archetype discovery

To interpret the learned risk structure, I fit a LightGBM model to the full residential dataset and compute SHAP values for all twelve features. SHAP values describe how each feature contributes to each parcel's predicted risk relative to the model baseline. I then cluster parcels in the twelve-dimensional SHAP space using k-means with $k = 4$.

Clustering in explanation space, rather than raw feature space, groups parcels according to how the model reasons about their risk. Each cluster is labeled from its SHAP centroid and then checked against geography and observed Harvey flood depths. Cluster stability is tested across random seeds and alternate k values; the selected $k = 4$ solution is highly stable across seeds.

3.4 Bootstrap uncertainty and counterfactual diagnostics

To quantify the impervious-surface relationship within each archetype, parcels are divided into ISA quintiles. For each archetype-by-quintile cell, I bootstrap mean flood depth 1,000 times with replacement. The main contrast is Q5 - Q1, or the difference between the most impervious and least impervious quintiles within the same archetype.

Finally, I run model-internal counterfactual diagnostics. In the primary dose-response analysis, each parcel's ISA is multiplied by factors from +20% to +100%, capped at 1.0, and Cw_topo is recomputed for consistency. The model then re-predicts risk classes. These counterfactuals are not physical simulations of land-cover change. They show how the trained model responds when ISA-like features are perturbed within the observed correlational structure.

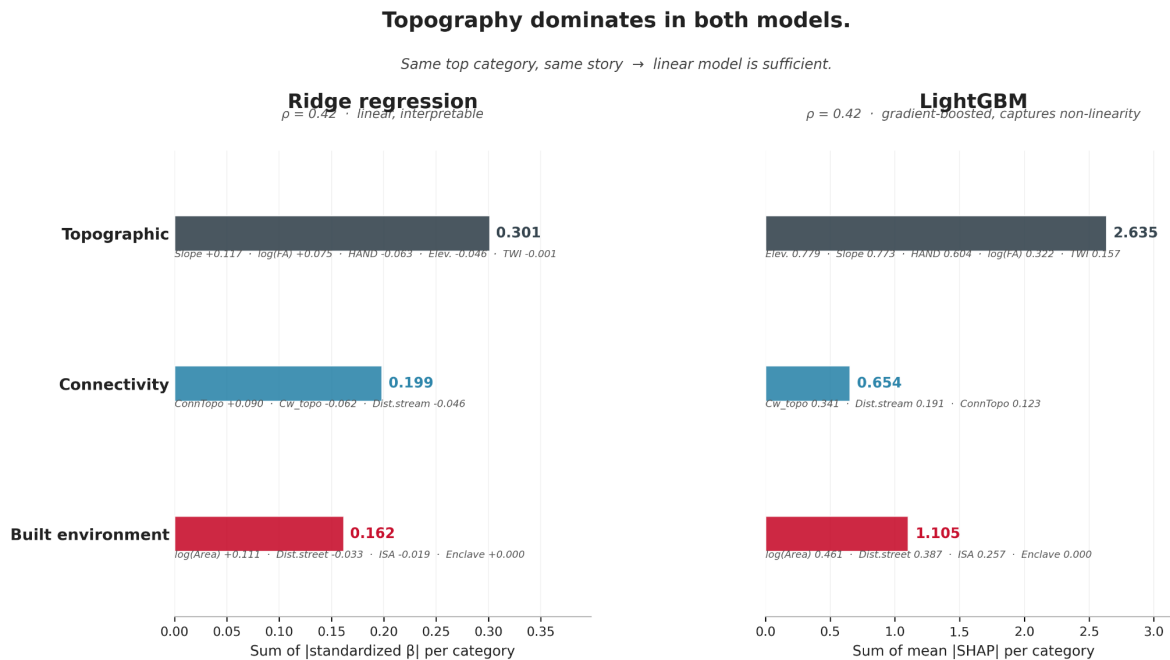


Figure 2. Topography dominates in both models. Ridge and LightGBM identify topographic predictors as the leading feature category, while Ridge provides a linear and interpretable headline model with comparable ranking performance.

4. Results

4.1 Ridge regression ranks parcel flood risk with useful accuracy

Under 2 km block cross-validation, the twelve-feature Ridge model achieves a Spearman rank correlation of 0.42 and NDCG@10% of 0.80 on the residential sample. This is well above the median baseline and above the single-

feature `Cw_topo` model, which reaches a Spearman value of approximately 0.30. LightGBM-LambdaRank performs similarly to Ridge, indicating that most of the predictive signal is captured by a relatively simple linear model.

Feature importance is consistent across Ridge and LightGBM. When features are grouped into topography, connectivity, and built environment, topographic predictors contribute the largest share of model importance in both approaches. Individual Ridge coefficients show positive associations for log flow accumulation and slope, negative association for distance to stream, and a small negative coefficient for `ISA_frac` after accounting for the other features. The enclave feature contributes no meaningful signal because no relevant enclave parcels remain after spatial filtering.

4.2 SHAP clustering reveals four residential flood-risk archetypes

Clustering the SHAP matrix produces four interpretable residential archetypes. The most important group is `Hotspot_res`, a small but highly vulnerable cluster representing 1.7% of residential parcels. These parcels are near drainage, have low `HAND_min`, high flow accumulation, and steep local slope signatures. Their mean Harvey flood depth is 3.27 ft, approximately 2.5 times the non-hotspot residential baseline.

The largest group is `Upland_shield_res`, representing 60.7% of residential parcels. These parcels occupy topographically protected positions and have a mean flood depth of 1.18 ft. The remaining two groups, `Lowland_baseline_res` and `Upland_baseline_res`, represent intermediate flood-risk regimes that lack the concentrated topographic signature of the hotspot class.

The archetypes are not only statistical clusters. Their geography follows the drainage structure of the watershed: hotspots trace the main channel and tributaries, while upland-shield parcels form the dominant residential matrix on higher ground. This spatial coherence supports the interpretation that the model has learned hydrologically meaningful topographic position rather than arbitrary feature combinations.

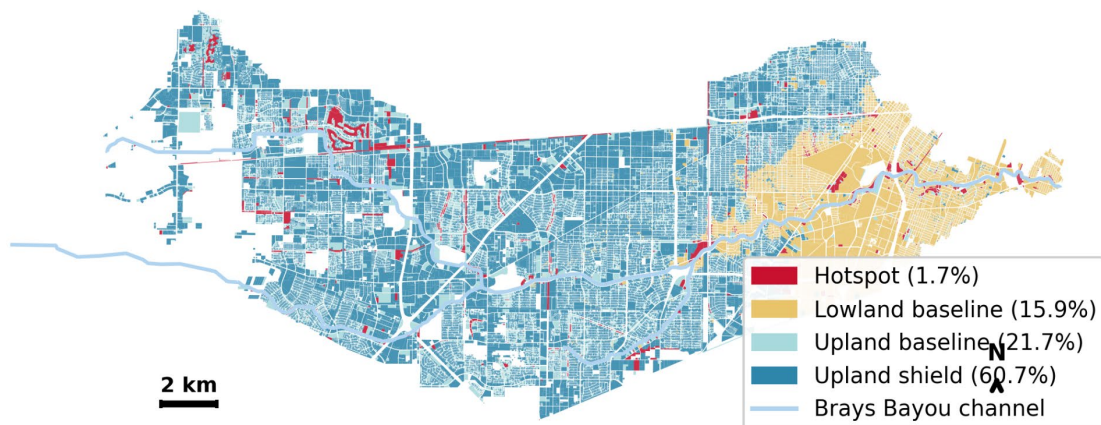


Figure 3. Residential flood-risk archetypes across Brays Bayou. SHAP clustering identifies four interpretable parcel types: Hotspot, Lowland baseline, Upland baseline, and Upland shield.

Table 2. Residential archetype summary.

Archetype	Share / n	Mean flood depth	Topographic signature	Interpretation
Hotspot_res	1.7%; n = 2,020	3.27 ft	Low HAND_min; near streams; high slope/flow accumulation	Small but extreme flood-depth regime
Upland_shield_res	60.7%; n = 70,252	1.18 ft	Elevated above nearest drainage	Dominant protected residential matrix
Lowland_baseline_res	15.9%; n = 18,471	1.59 ft	Low-lying but not hotspot-convergent	Moderate exposure
Upland_baseline_res	21.7%; n = 25,077	1.41 ft	Moderately elevated	Intermediate/protected exposure

4.3 The impervious paradox appears within every archetype

The central empirical result is that ISA is negatively associated with observed Harvey flood depth within every residential archetype. In each group, the most impervious quintile flooded less than the least impervious quintile. The inversion is strongest in Hotspot_res: the most impervious hotspot parcels flooded 1.31 ft less than the least impervious hotspot parcels, with a 95% bootstrap confidence interval excluding zero.

The effect is smaller but still consistent in the other three archetypes: -0.18 ft in Lowland_baseline_res, -0.19 ft in Upland_baseline_res, and -0.12 ft in Upland_shield_res. All confidence intervals exclude zero. The direction of the relationship is therefore not driven by one unusual cluster; it is repeated across the residential flood-risk landscape.

This result should not be interpreted as evidence that paving physically protects parcels from flooding. Instead, it shows that in this mature urban watershed, ISA is entangled with historical siting. Less impervious parcels are disproportionately located in low-lying or drainage-adjacent positions, while highly impervious residential parcels often occupy already-developed and topographically safer land. The observed parcel-scale ISA signal therefore carries both hydrologic and historical-geographic information.

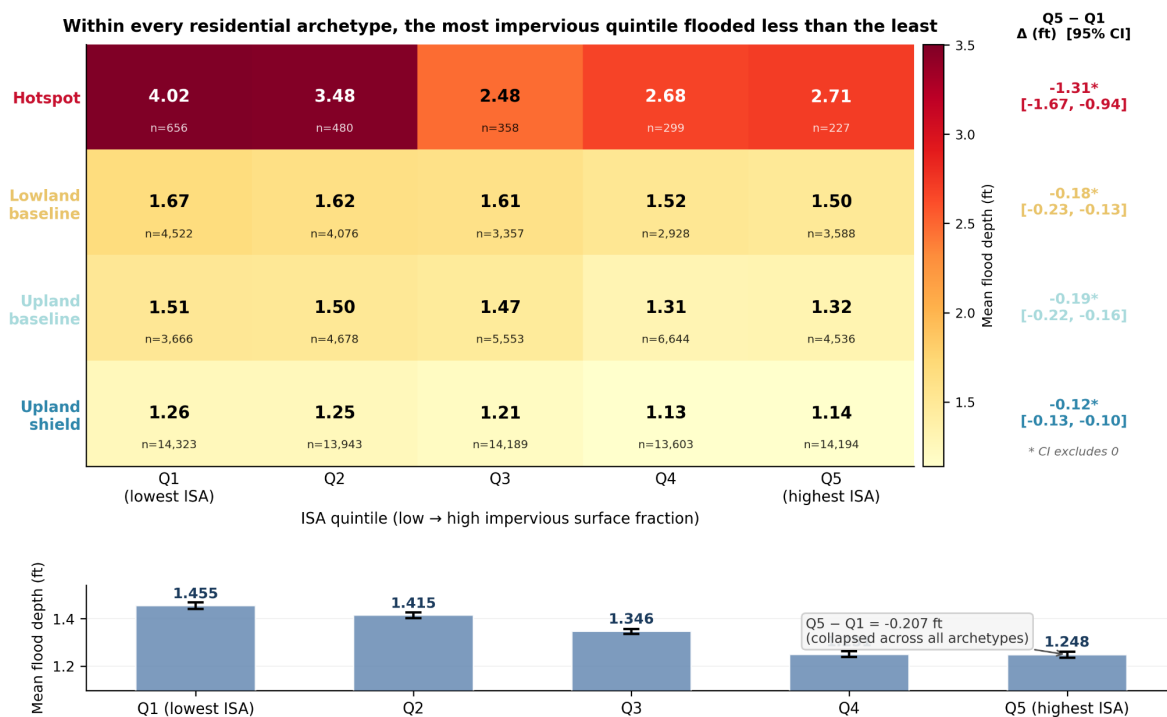


Figure 4. Impervious paradox by residential archetype. Within every archetype, the most impervious quintile flooded less than the least impervious quintile during Hurricane Harvey; the inversion is strongest in topographic hotspots. The right-side column reports the Q5 minus Q1 difference with 95 percent confidence intervals.

Table 3. Q5 - Q1 impervious-surface contrast by archetype.

Archetype	Q5 - Q1 flood depth (ft)	95% bootstrap CI
Hotspot_res	-1.31	[-1.67, -0.94]
Lowland_baseline_res	-0.18	[-0.23, -0.13]
Upland_baseline_res	-0.19	[-0.22, -0.16]
Upland_shield_res	-0.12	[-0.13, -0.10]

4.4 Simulated densification recruits moderate risk but barely expands catastrophic risk

The dose-response analysis perturbs ISA from +20% to +100% and tracks movement across predicted risk classes. The main pattern is saturation: at +20% ISA, 8.6% of residential parcels jump at least one risk class; at +40%, roughly 10% have jumped; after +60%, the curve becomes nearly flat. By that point, many parcels have reached the ISA = 1.0 ceiling and cannot respond further in the model.

Risk recruitment is concentrated in currently protected parcels rather than already-vulnerable hotspots. At +20% ISA, approximately 73% of class-jumping parcels come from Upland_shield_res, the safest and largest archetype at baseline. Hotspot_res shows a higher continuous-risk response but a lower class-jump rate because many hotspot parcels are already near the highest risk class. This is a ceiling effect: development can change moderate-risk status more easily than it can expand the set of parcels capable of catastrophic flooding under Harvey-like conditions.

Across all densification scenarios, only a very small number of parcels newly enter the highest risk class. This suggests that the catastrophic-risk frontier is primarily topographically defined, while additional imperviousness mainly shifts parcels within the moderate-risk distribution. Within the limits of the model-internal counterfactual, the timing implication is that flood-risk management should pay attention to early densification of currently protected upland parcels, before they move into the saturation regime.

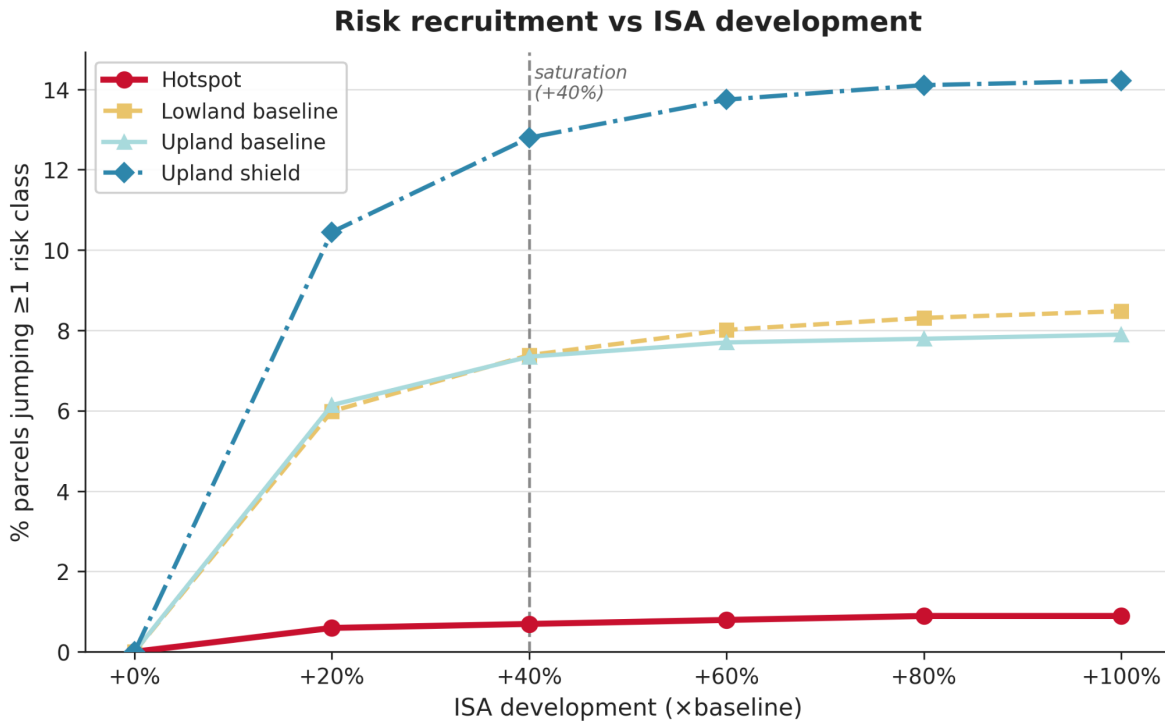


Figure 5. *Densification dose-response. Model-internal ISA perturbations recruit parcels into higher risk classes, but the response saturates near +40 percent ISA and is driven primarily by parcels currently in the Upland shield archetype.*

5. Discussion

5.1 Interpreting the impervious paradox

The impervious paradox does not overturn watershed hydrology. At the watershed scale, impervious surface can increase runoff and worsen downstream flooding. The contribution of this paper is narrower: at the parcel scale, within an already urbanized and low-relief watershed, ISA does not necessarily identify the parcels that flooded most deeply during Harvey.

The most plausible explanation is spatial confounding. In mature watersheds, low-ISA parcels are often not randomly distributed. They include floodplain reserves, drainage easements, parks, large lots near streams, and land that remained less developed partly because it was more difficult or risky to build on. High-ISA residential parcels, by contrast, often represent land that has already been selected, subdivided, serviced, and developed. In this setting, ISA partly encodes historical siting decisions.

The model therefore learns two signals at once. One is the physical relationship between imperviousness, connectivity, and runoff. The other is the historical relationship between land cover and topographic position. At parcel scale in Brays Bayou, the siting proxy appears strong enough that the observed ISA-flood relationship becomes negative within every residential archetype.

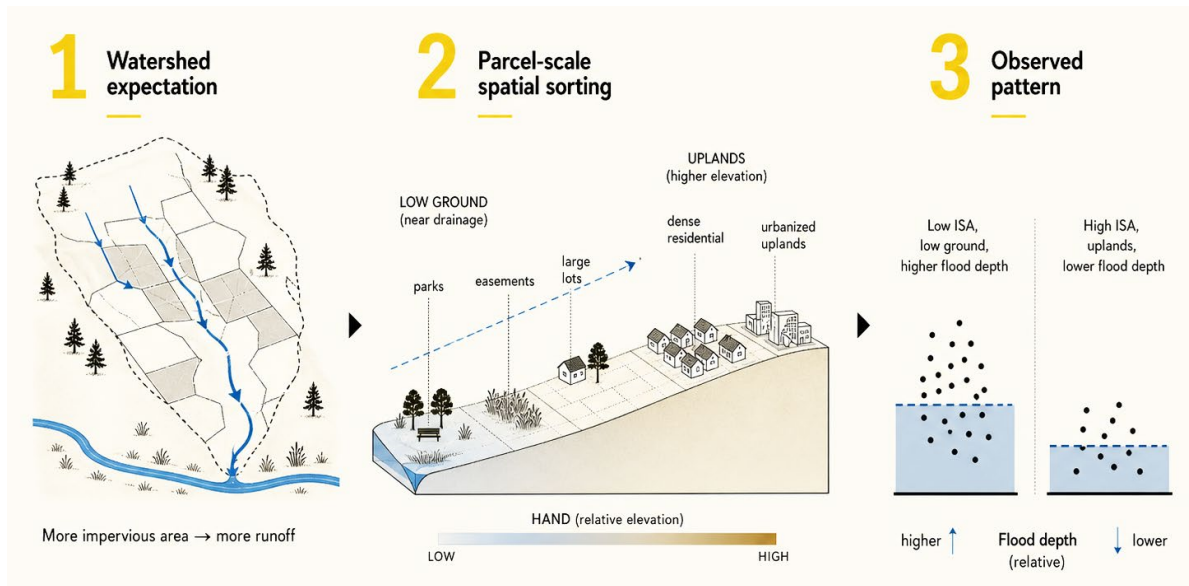


Figure 6. *Conceptual mechanism of the impervious paradox. The diagram contrasts the watershed-scale expectation that more imperviousness increases runoff with the parcel-scale reality that spatial sorting places low-ISA parcels in lower ground and high-ISA parcels in better-drained uplands, producing the observed reversal in flood depth.*

5.2 Causal limits of the counterfactual analysis

The counterfactual results should be read carefully. They are model-internal feature perturbations, not hydraulic simulations and not predictions of what would physically happen if a parcel were depaved, paved, or retrofitted. Reducing ISA in the trained model moves a parcel toward the statistical profile of historically low-ISA parcels. If historically low-ISA parcels were disproportionately flood-prone, the model can predict higher risk after an ISA reduction even though physical depaving may reduce runoff at the site.

This distinction is especially important because Cw_topo is algebraically linked to ISA. Ridge regression can assign different effective roles to ISA_frac and Cw_topo because they represent different aspects of the joint feature distribution. The counterfactuals therefore diagnose what the model has learned; they do not replace hydraulic modeling, controlled intervention studies, or post-implementation monitoring.

A useful next step would be to pair this parcel-ranking model with physically based hydrologic or hydraulic simulations. The machine-learning model can identify where risk concentrates and how predictors behave statistically. A hydraulic model can test whether specific interventions would reduce depth, duration, or velocity under particular storm scenarios.

5.3 Design and planning implications

For parcel-scale flood screening, the results suggest that topographic position should be the first-order signal. Parcels with low $HAND_min$, high flow accumulation, stream proximity, and convergent local slope deserve attention even when they are not highly impervious. Conversely, high imperviousness alone should not be treated as evidence of high parcel flood depth without checking the parcel's drainage position.

For urban design, the finding changes the intervention question. Instead of asking only where to reduce impervious surface, planners should ask where the watershed's flood-storage and flow-convergence functions are located, who occupies those locations, and whether future development would remove informal flood-storage capacity. This is especially relevant for public/green parcels that may quietly absorb floodwater but are not always valued as flood infrastructure.

For development policy, the dose-response result suggests that early densification may matter more than late densification in already saturated watersheds. Once a landscape is highly impervious, additional paving may have limited marginal effect on model-predicted class transitions. The more important planning window may be the period when currently protected upland parcels begin moving from low to moderate risk.

5.4 Limitations

This study has five main limitations. First, it analyzes one event: Hurricane Harvey. Harvey was an extreme, long-duration, saturation-limited storm. The ISA relationship may differ in shorter, high-intensity storms where infiltration capacity and local drainage performance play different roles.

Second, the analysis focuses on one Houston watershed. Brays Bayou's flat coastal topography, development history, and drainage infrastructure may produce a stronger siting confound than other cities. Replication in other mature urban watersheds, including New Orleans, Miami, Rotterdam, and Mumbai, would test generality.

Third, the data are cross-sectional. A temporal analysis using multiple storms and historical ISA layers would better separate land-cover change from inherited siting. Fourth, the model's Spearman correlation of 0.42 is useful but moderate; parcel-scale flood depth also depends on stormwater infrastructure, building modifications, fences and walls, sub-10 m microtopography, and local drainage failures not captured in the twelve-feature set.

Fifth, the archetypes are model-dependent. SHAP clusters summarize how this LightGBM model explains risk, not intrinsic natural classes of urban land. Their stability across seeds supports their use as interpretive summaries, but different models or targets could produce different archetype boundaries.

6. Conclusion

This report tested whether the standard urban flood-design prescription--reduce impervious surface to reduce flood risk--holds at the parcel scale in Houston's Brays Bayou watershed. Using 115,820 residential parcels, Hurricane Harvey flood depths, spatial cross-validation, Ridge regression, and SHAP-based archetype discovery, I find that topographic position dominates parcel-scale flood-risk prediction.

The central result is an impervious paradox: within every residential archetype, the most impervious parcels flooded less than the least impervious parcels. This does not mean impervious surface physically prevents flooding. It means that in a mature urban watershed, imperviousness is entangled with historical siting. Low-ISA land is often located in flood-prone positions, while heavily developed parcels may occupy topographically safer ground.

For parcel-scale flood planning, the implication is to begin with drainage geometry: $HAND$, flow accumulation, stream proximity, slope, and local connectivity. Surface cover still matters, but it should be interpreted as a context-dependent modifier rather than a universal risk proxy. The methodological lesson is broader: predictive variables in

Earth and urban systems can represent physical causes, spatial proxies, or both. The challenge is not only to predict risk, but to understand what kind of signal the model has learned.

References

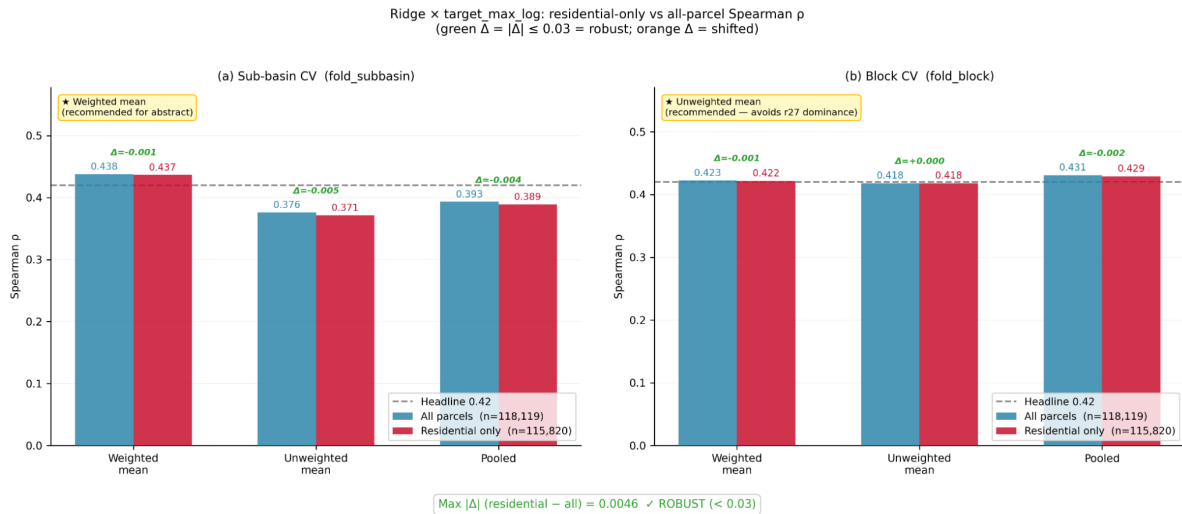
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Appendix: Material to keep outside the main 6-8 page report

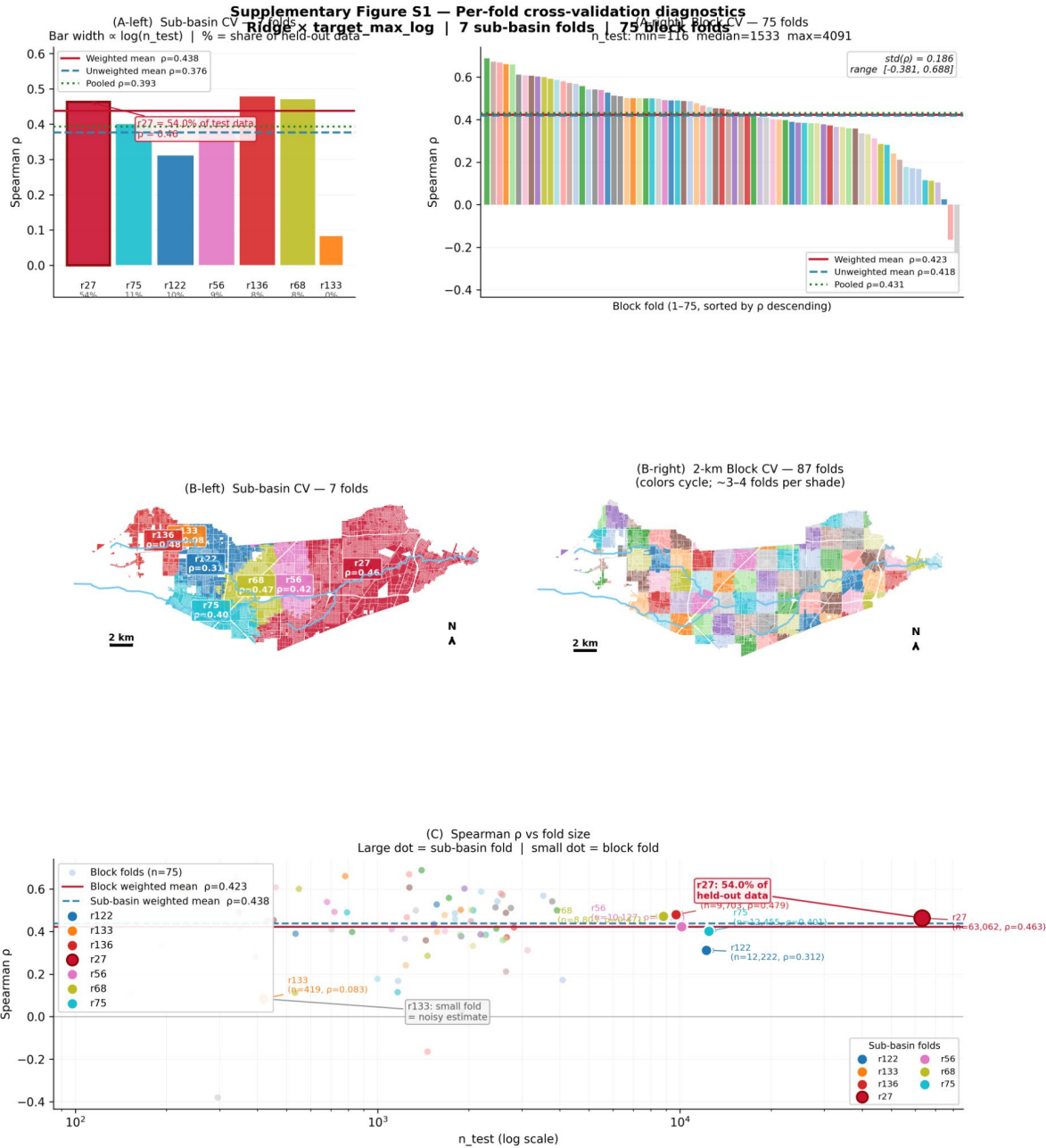
The supplementary figures below document robustness checks, cross-validation structure, and fold-wise diagnostics that support the main-text results.

A1. Full hyperparameters: Ridge alpha, LightGBM parameters, Random Forest parameters, k-means settings, software versions, and preprocessing details.

A2. Per-fold cross-validation diagnostics: sub-basin and 2 km block fold maps, per-fold Spearman values, fold-size sensitivity, and pooled/weighted/unweighted comparisons.

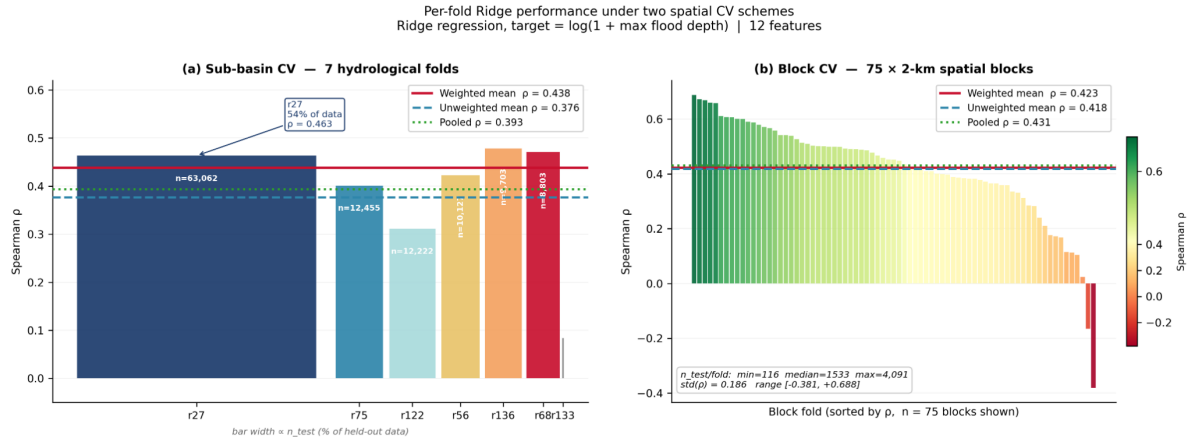


Supplementary Figure S1. Residential-only versus all-parcel robustness. Under both sub-basin and block cross-validation, Ridge performance changes only minimally when the dataset is restricted to residential parcels, indicating that the headline result is robust to parcel filtering.



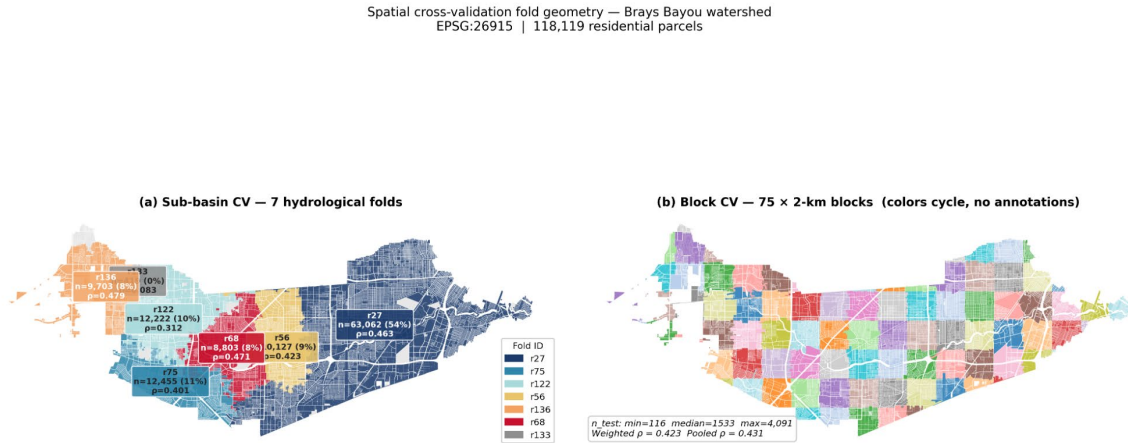
Supplementary Figure S2. Per-fold cross-validation diagnostics. Fold-wise model performance is shown for both sub-basin and block cross-validation, together with fold maps and the relationship between Spearman rho and held-out fold size.

A3. Cluster stability: ARI matrix across k values and random seeds, plus rationale for choosing k = 4.



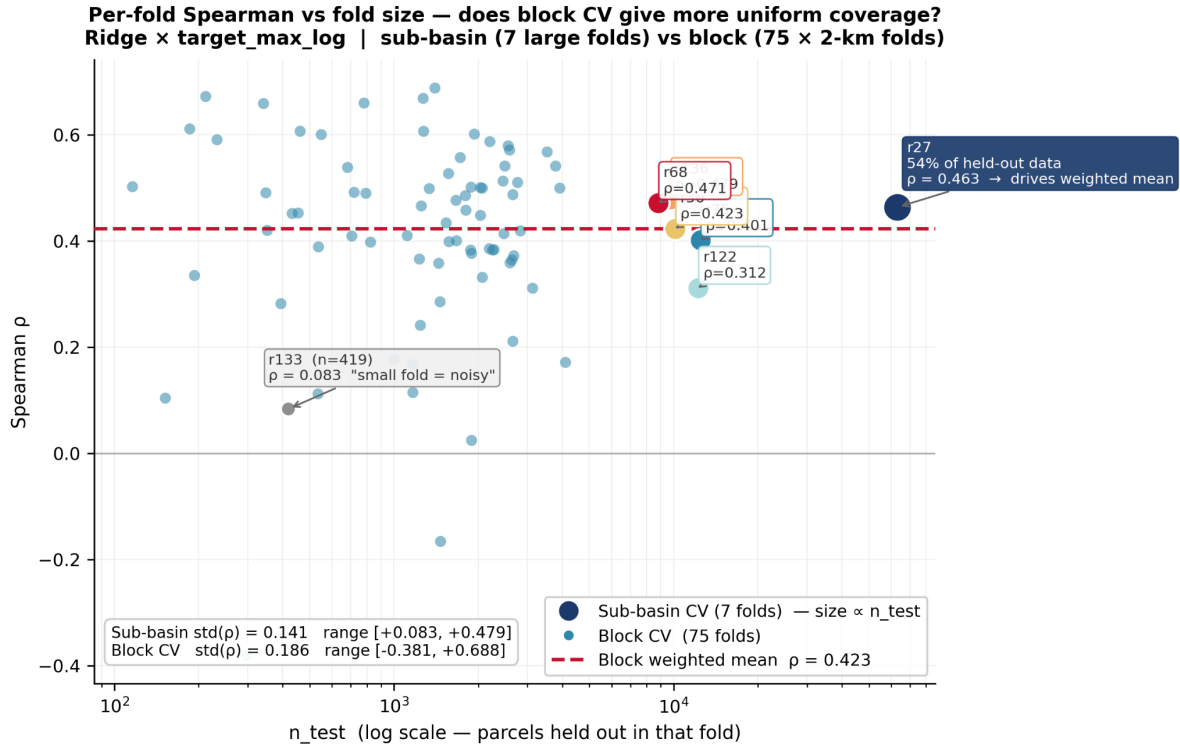
Supplementary Figure S3. Ridge performance under two spatial cross-validation schemes. Sub-basin CV yields a weighted mean Spearman rho of 0.438, while 2 km block CV yields 0.423, supporting the paper’s headline generalization accuracy.

A4. Flood-depth distributions by archetype: violin plots capped at the 97th percentile, with medians and interquartile ranges.



Supplementary Figure S4. Spatial cross-validation fold geometry. The left panel shows the seven hydrologic sub-basin folds, and the right panel shows the 75 two-kilometer block folds used for the stricter spatial generalization test.

A5. Intervention diagnostic: depave, permeable paving, and bioswale model-internal perturbations. Keep a clear warning that these are diagnostic results, not physical prescriptions.



Supplementary Figure S5. Spearman rho versus held-out fold size. Block CV provides more uniform spatial coverage, but the larger number of small held-out folds contributes to wider fold-to-fold variability.

A6. Reproducibility: Code, processing scripts, summary outputs, and figure-generation materials are available at the project repository: <https://github.com/lorenzo1496/impervious-paradox>. The repository serves as the companion code archive for this AI for Earth and Planetary Science submission.